

NPTEL Python Course - Multiple Choice Questions (MCQs)

Week 2: Python Basics

1. What will be the output of `bool([])` in Python?

- a) True b) False c) None d) Error

Answer: b) False

2. Which function is used to get user input in Python?

- a) `get()` b) `input()` c) `read()` d) `scanf()`

Answer: b) `input()`

3. What will `print(type(5/2))` output?

- a) `<class 'int'>` b) `<class 'float'>` c) `5/2` d) Error

Answer: b) `<class 'float'>`

Week 3: Lists, Loops, and Functions

11. What is the output of `list(range(2, 10, 2))`?

- a) `[2, 3, 4, 5, 6, 7, 8, 9]` b) `[2, 4, 6, 8]` c) `[2, 4, 6, 8, 10]` d) None

Answer: b) `[2, 4, 6, 8]`

12. What does `list1.append(4)` do?

- a) Adds 4 to the beginning of list1 b) Adds 4 to the end of list1 c) Removes 4 from list1 d) Sorts list1

Answer: b) Adds 4 to the end of list1

Week 4: Sorting Algorithms

21. What is the worst-case time complexity of selection sort?

- a) $O(n \log n)$ b) $O(n^2)$ c) $O(n)$ d) $O(1)$

Answer: b) $O(n^2)$

22. Merge sort follows which paradigm?

- a) Greedy b) Divide and conquer c) Dynamic programming d) Brute force

Answer: b) Divide and conquer

Week 5: File Handling and Errors

31. What function is used to open a file in Python?

a) open() b) file.open() c) openfile() d) readfile()

Answer: a) open()

32. What will happen if you try to open a non-existent file in "r" mode?

a) Creates a new file b) Returns None c) Raises a FileNotFoundError d) Opens an empty file

Answer: c) Raises a FileNotFoundError

Week 6: Backtracking and Data Structures

41. What type of algorithm is used to solve the N-Queens problem?

a) Greedy Algorithm b) Backtracking c) Dynamic Programming d) Divide and Conquer

Answer: b) Backtracking

42. Which data structure is used to implement backtracking?

a) Queue b) Stack c) Linked List d) Hash Table

Answer: b) Stack

Week 7: Binary Search Trees

51. In a Binary Search Tree (BST), where are values smaller than the root stored?

a) Left subtree b) Right subtree c) Both left and right subtrees d) Nowhere

Answer: a) Left subtree

52. What is the best-case time complexity of searching for an element in a BST?

a) $O(n \log n)$ b) $O(\log n)$ c) $O(1)$ d) $O(n^2)$

Answer: c) $O(1)$, when the searched element is at the root

Week 8: Dynamic Programming and Matrix Multiplication

61. What is the time complexity of multiplying two $n \times n$ matrices using the standard method?

a) $O(n^3)$ b) $O(n \log n)$ c) $O(n^{2.373})$ d) $O(n^2)$

Answer: a) $O(n^3)$

62. What technique is used to find the longest common subsequence (LCS) between two strings?

a) Divide and Conquer b) Backtracking c) Dynamic Programming d) Brute Force

Answer: c) Dynamic Programming